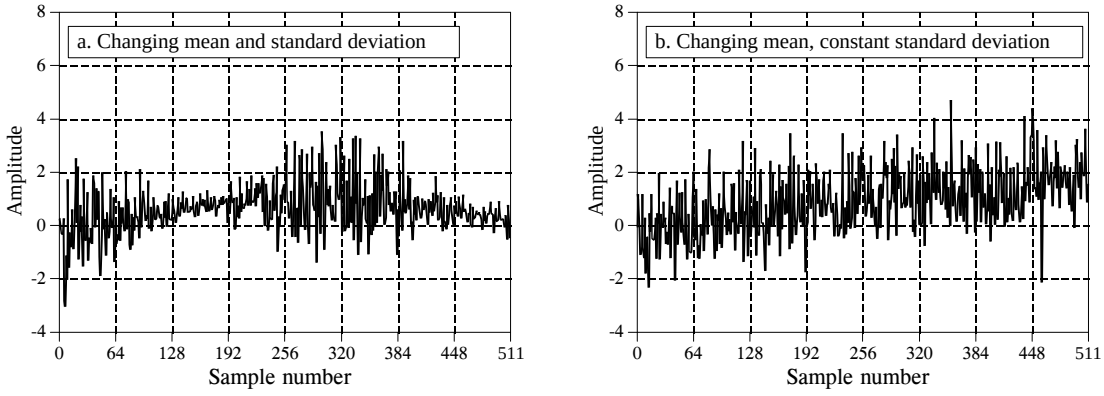

FIGURE 2-3
Examples of signals generated from nonstationary processes. In (a), both the mean and standard deviation change. In (b), the standard deviation remains a constant value of one, while the mean changes from a value of zero to two. It is a common analysis technique to break these signals into short segments, and calculate the statistics of each segment individually.

estimate of the standard deviation of the underlying process. In other words, Eq. 2-2 is an *estimate* of the standard deviation of the *underlying process*. If we divided by *N* in the equation, it would provide the standard deviation of the *acquired signal*.

As an illustration of these ideas, look at the signals in Fig. 2-3, and ask: are the variations in these signals a result of statistical noise, or is the underlying process changing? It probably isn't hard to convince yourself that these changes are too large for random chance, and must be related to the underlying process. Processes that change their characteristics in this manner are called **nonstationary**. In comparison, the signals previously presented in Fig. 2-1 were generated from a stationary process, and the variations result completely from statistical noise. Figure 2-3b illustrates a common problem with nonstationary signals: the slowly changing *mean* interferes with the calculation of the *standard deviation*. In this example, the standard deviation of the signal, over a short interval, is *one*. However, the standard deviation of the entire signal is 1.16. This error can be nearly eliminated by breaking the signal into short sections, and calculating the statistics for each section individually. If needed, the standard deviations for each of the sections can be averaged to produce a single value.

The Histogram, Pmf and Pdf

Suppose we attach an 8 bit analog-to-digital converter to a computer, and acquire 256,000 samples of some signal. As an example, Fig. 2-4a shows 128 samples that might be a part of this data set. The value of each sample will be one of 256 possibilities, 0 through 255. The **histogram** displays the *number of samples* there are in the signal that have each of these *possible values*. Figure (b) shows the histogram for the 128 samples in (a). For